

How many more cars would Prague's paid kerb hold with 2012's car lengths?

On a fleet model, today's paid kerb in Prague would hold about 1.6 % more cars of 2012's lengths, and new Czech cars grew less over 2022–2025 than T&E's series implies (supported).

MORE CARS TODAY'S KERB WOULD HOLD WITH 2012'S CARS	1.6 % 1.3–1.9 % · range under the registered inputs
A NEW CAR IN CZECHIA, 2022 → 2025	+2.3 cm 95 % CI -0.3 to +5.2 · bootstrap by model line
STALLS IN THE PAID ZONES, 28 SEPTEMBER 2026	168 424 administrative count
A LONG CAR'S ADVANTAGE UNDER THE FLAT PERMIT	~120 Kč a year, revenue-neutral length price · 90th vs 10th length percentile · 114–124 Kč across the registered gap range

OVERWHELMED? HERE'S THE SHORT VERSION

- The kerb figure is a counterfactual on a fleet model, not a count; per space on parallel kerb the gain is larger (2.7 %), and within the registered input ranges almost all of its spread comes from how long cars made before 2010 really were.
- Growth of new cars over 2022–2025 passed the registered test against T&E's 6 cm, though no growth at all is not ruled out; whether the register agrees with the earlier wheelbase route is inconclusive (underpowered).
- Painted bays could only lower the estimate and a gap proportional to car length would raise it; the modelled 2025 fleet is 3.9 cm shorter than the register's, a failed calibration check, and five registered robustness checks were not run.
- The flat resident permit charges a short and a long car the same, and it costs far less than the city's own second-car price; both are policy prices, not rents.

- It does not show how many stalls Prague had in 2012, how drivers would respond to a different price, or how the difference compares with the growth in the number of cars.

published 14 September 2026 (first estimate; this study 29 September 2026) · updated 9 October 2026 · version 3

work since May 2024

+ About this study



The question

Cars sold in Europe have been getting longer and wider, a change Transport & Environment (T&E) has measured in the average dimensions of new cars sold in the EU, the UK and Norway¹. Where cars park end to end, a longer car leaves less room for the next one. This study asks how much more of Prague's paid kerb today's cars take than 2012's cars would, and what the flat resident permit charges for the difference. Parking economics has long treated the kerb as underpriced land whose price, more than its supply, rations its use^{2,3,4,5}, and T&E's projection that European cities lose 8.5 to 14 % of parallel parking by 2040 has brought the size of cars into the same argument¹.

Three earlier texts on this site answered the question with numbers they could not check^{6,7,8}: the growth of new cars from T&E's European series, the share of parallel parking from Berlin, and a length series converted from EEA wheelbases. This study replaces them. It reads the length of every car in the Czech vehicle register and the geometry of every segment of Prague's paid zones, under an analysis plan committed before this analysis was run.

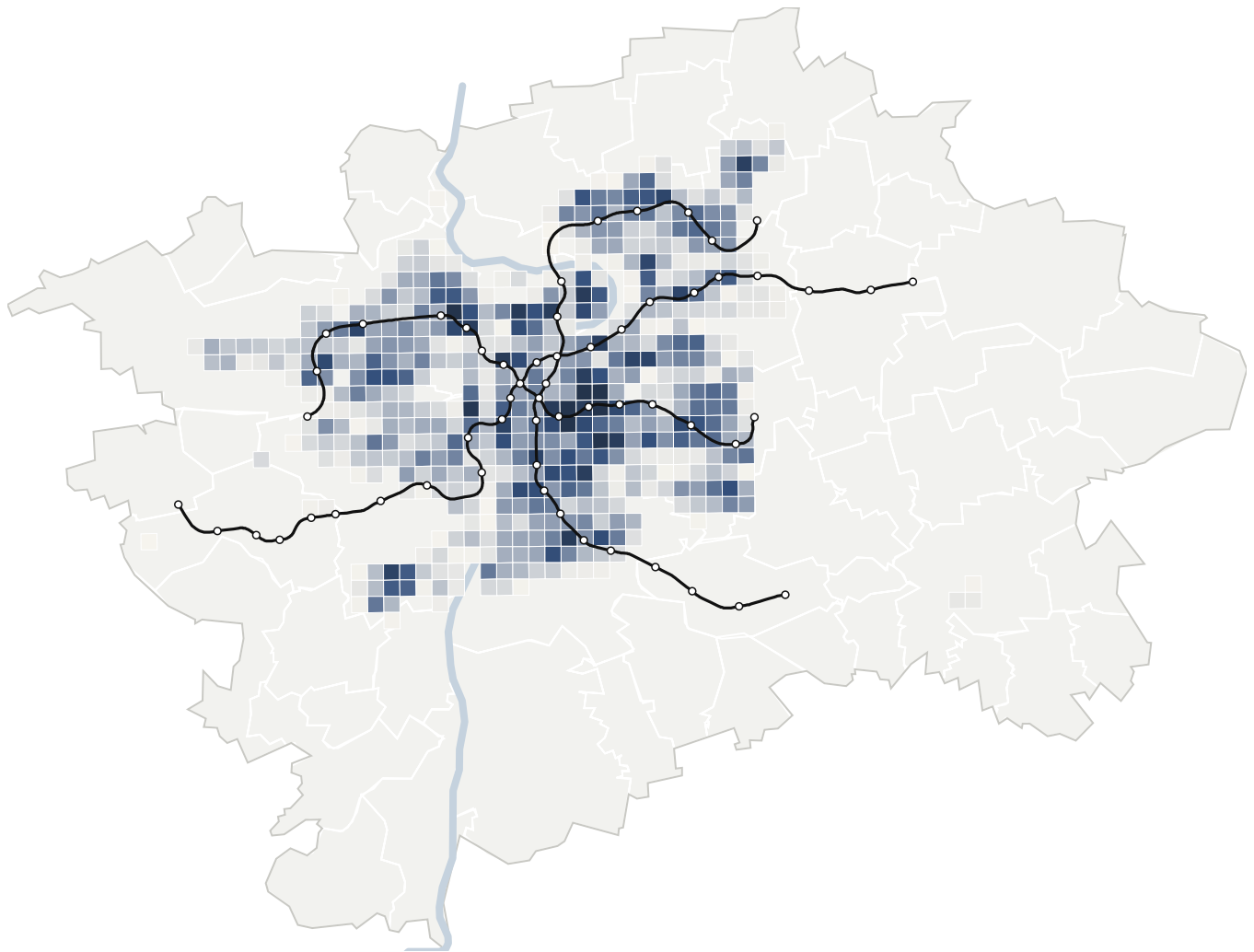
Data

The size of the cars comes from the Ministry of Transport's register, which lists the vehicles registered in Czechia, including those since scrapped or exported, with their overall length². The export of 1 September 2026 has 19.4

million rows. Passenger cars and their off-road variant (M1 and M1G) number 13.4 million, of which 6.9 million are in operation. Length is recorded for at least 99.6 % of cars made since 2012, but for fewer than half of those made before 1990. New cars in each year are weighted to the EEA's registration counts model line by model line (a model line is the make plus the first word of the model name, such as Škoda Octavia), so that cars since scrapped or exported count as they did when they were sold¹⁰. The EEA files carry wheelbase but not length, and no dimensions at all for Czech cars after 2022; how they were cleaned is set out in the design file¹¹ and the archived wheelbase text⁷.

The kerb comes from the city's layer of paid parking zones on its ArcGIS server, where each segment has an area in real square metres¹². The server works in S-JTSK, the Czech national grid, so nothing is reprojected, whereas the copy on ArcGIS Online is in Web Mercator and overstates every area by a factor of 2.43 at Prague's latitude. On 28 September 2026 the layer had 16 874 segments holding 168 424 stalls, 1.4 % above the count that TSK, the city's road authority, published for December 2024¹³, which is the expected direction since capacity has been added since then. These are administrative stalls, a count the city assigns to each segment.

Whether a segment is parallel or perpendicular can be read from its shape, since a band 2 to 2.5 m wide suggests cars along the kerb and one 4.5 to 5.5 m wide cars across it. This study reads the band width of each segment from its area and perimeter. Checked against 1 706 segments that the city had labelled in 2019, all in Prague 4, 5, 6 and 9, the rule classifies 97 % of stalls correctly. Applied to today's layer, it puts 59 % of stalls on parallel kerb, where the earlier texts had borrowed 68 % from Berlin¹.



0 1 000+ · stalls · per 450 m cell, up to 1 261

Hatched: cells with fewer than 10 stalls, where the ratios are left blank. Left off the map: Praha 16, 18 stalls.

base: Prague's city districts from the election precincts; river © OpenStreetMap contributors (ODbL)

map · Prague's paid parking zones on 28 September 2026: the city's 16 874 segments, holding 168 424 administrative stalls, summed into 548 cells of a 450 m grid (each segment counted in the cell that holds its centre); the one cell in Praha 16 (Radotín, 18 stalls), 4 km south of the rest, is left off, so the map shows 547. Pick a view above the map (stalls, gross zone area per stall, the share of stalls on parallel kerb by the band-width rule, or the share in resident, mixed and visitor segments) and hover or tap a cell for all its numbers. Hatched: cells with fewer than 10 stalls, where the ratios are left blank. Black: the metro and its stations. Source: the city's paid parking zone layer¹², snapshot of 28 September 2026.

How this was done

The study follows a registered analysis plan: design committed at a1ea422 on 29 September 2026, before the results commit 497925a; commit times are self-reported, and the branch was pushed after the analysis¹¹. The author has worked with these sources, the register among them, continuously since May 2024, in work that led to this study, so the plan was written with knowledge of the sources, not blind to them; it sets the analysis in advance. The code was then frozen and run once on lengths scrambled across years within each make, and only after that on the real ones. The plan fixes two tests, which share a 5 % family-wise error rate (the chance that either test wrongly passes) through Holm's procedure¹⁴, and three estimates that are reported rather than

tested, each with bands (ranges such as 1–2.5 %) and a reporting sentence for each band written in advance. It also fixes the inputs and their ranges.

On a kerb where cars park end to end, capacity is the length of kerb divided by the pitch, the car's length L plus the gap g between cars. Width does not enter, by assumption: the model assumes that a wider car takes its extra width from the driving lane and not from the cars in front and behind. The city publishes no kerb length, so the model holds today's kerb fixed and anchors on today's count: the stalls a past fleet would fill equal today's stalls times the ratio of the two pitches,

$$N_{2012} = N_{2025} \cdot \frac{L_{2025} + g}{L_{2012} + g}, \quad g = p - L_{2025}$$

where p is the end-to-end pitch, drawn between 4.9 and 5.5 m around T&E's 5.20 m from Berlin traffic data¹, and L_{2025} the modelled fleet mean of 2025, so that the gap is about 0.83 m (0.59–1.06 m).

New lengths reach the parked fleet slowly, because the average car in the Czech fleet is almost seventeen years old¹⁵. The fleet is modelled from the register. Every car entering in each year, at the age it entered (new, or imported used), survives to 2012 and to 2025 on a Weibull survival curve calibrated to the fleet's mean age in each year, 16.7 years in 2025, which is published, and an assumed 13.3–14.1 years in 2012, which is not. The intervals of the three estimates are the 2.5–97.5 % range over draws from the registered input distributions, so they describe uncertainty in the inputs and are not sampling intervals. The tests use a studentised bootstrap clustered by model line, which resamples whole model lines so that cars of one model are not treated as independent.

The tests

Hypothesis	Test	Result
H2a · the register agrees with the wheelbase estimate on how much new cars grew, 2012–2022	equivalence test: is the difference shown to lie within ±4.4 cm? ¹⁶	inconclusive (underpowered) +2.3 cm apart, p = 0.29
H2b · new cars grew less in 2022–2025 than T&E's series implies (+6.0 cm)	one-sided	supported +2.3 cm, p = 0.007

p-values: studentised cluster bootstrap over 796 model lines, 9 999 draws. Holm thresholds 0.025 and 0.05; only H2b passes. The ±4.4 cm margin is the change in new-car growth that would move the per-parallel-space estimate by a quarter of a percentage point, fixed from constants before this analysis was run.

New cars in the register, 2012–2025

Measured from the register and weighted to the EEA's counts, the average new car in Czechia grew by 12.7 cm per decade over 2012–2022. H2a compares that with the route the earlier wheelbase text took, which turned the EEA's wheelbase series into length⁷. With the off-road variant included and wheelbase converted at the register's own ratio within each model line, 1.47 mm of length per mm of wheelbase, that route gives 10.4 cm, while the published European ratio, wheelbase 0.618 of length¹⁷ or 1.62 mm of length per mm of wheelbase, would turn the same wheelbase trend into 11.4 cm. The register and the wheelbase route differ by 2.3 cm, and the test cannot say whether that counts as agreement, because the 90 % CI runs from –4.3 to +9.2 cm, far wider than the

margin. The plan's power check, run before this analysis, had put the chance of showing agreement at 16 %, and the realised standard error was 3.9 cm against the 2.4 cm planned, so the real power was lower still. The result is inconclusive (underpowered) and does not indicate that the two routes disagree.

New cars in Czechia grew 2.3 cm over 2022–2025, less than the 6 cm that T&E's series implies.

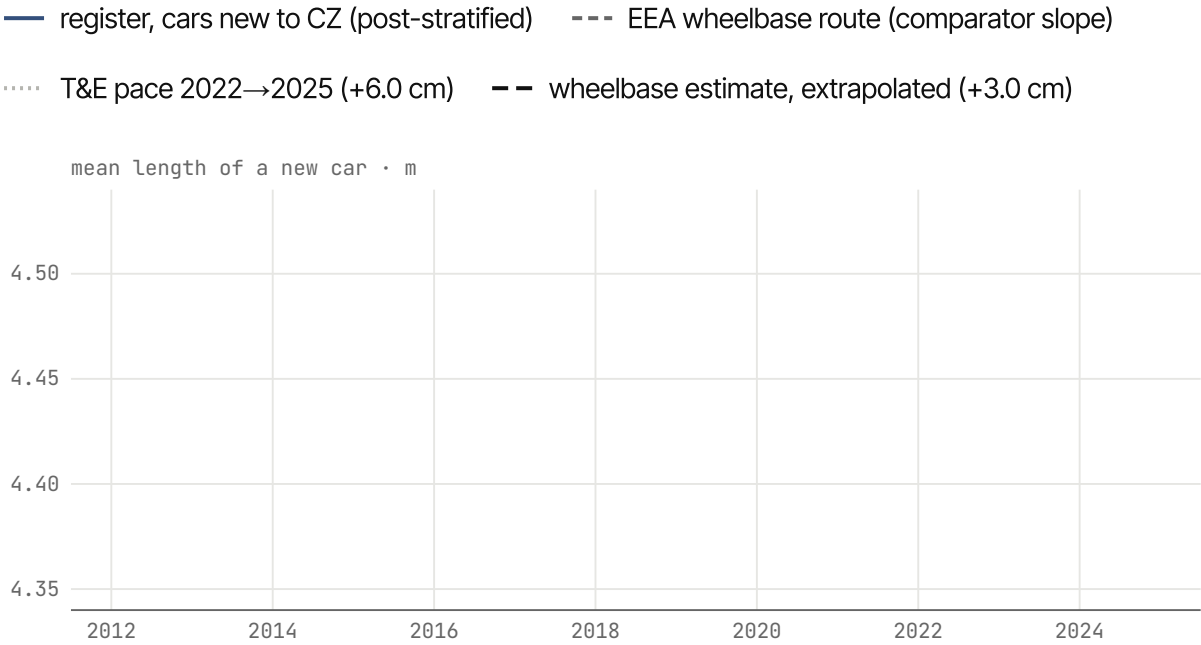


fig. 1 · Mean overall length of cars new to Czechia, in metres, from the register, weighted to EEA registrations by model line (blue, with dots). Dashed grey: the slope of the wheelbase route, the H2a comparator. After 2022, both starting from the register's 2022 value: T&E's pace (dotted light grey) and the wheelbase estimate's extrapolation (long-dashed black). Sources: vehicle register, export of 1 September 2026; EEA, final files 2010–2022 and counts 2023–2025.

For 2022 to 2025, the years for which the EEA publishes no dimensions, the register gives a firmer result. New cars changed by +2.3 cm over those three years (95 % CI –0.3 to +5.2, bootstrap by model line), against the 6.0 cm that T&E's European series implies¹, and the registered test against 6.0 passes ($p = 0.007$). The test asks whether growth was below 6 cm, not whether it was above zero: the interval reaching –0.3 cm means the data cannot rule out that new cars did not grow at all, while they do rule out 6 cm. The plan's other question, whether the Czech market kept the wheelbase estimate's pace of about 3 cm, cannot be answered from these data, since the interval covers both 3 and the midpoint of 4.5. T&E's figures are rounded to the centimetre, and since its 2022 value is interpolated between 2020 and 2025, the rounding moves the three-year difference by up to about 0.6 cm. In the registered sensitivity runs H2b stays at 2.1–2.7 cm; leaving out the Octavia, 10 % of the weight, moves the H2a difference to 4.2 cm, and endpoint means instead of the trend give 4.0 cm.

One registered test passes, growth below 6 cm; whether the two routes agree is too imprecise to say.

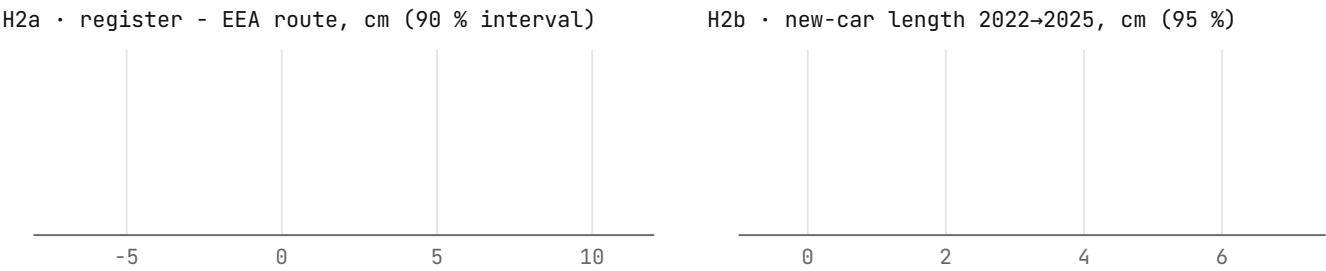


fig. 2 · The two registered tests, in centimetres, bootstrap by model line (796 lines). Left: H2a, the difference between the two routes with the 90 % CI an equivalence test at 5 % uses, against the ±4.4 cm margin (blue band). Right, or below on narrow screens: H2b, three-year growth with its 95 % CI, against the wheelbase estimate's 3.0 cm (dashed grey), the midpoint (dashed light grey) and T&E's 6.0 cm (dashed black). Source: vehicle register, export of 1 September 2026.

What today's kerb would hold with 2012's cars

On the fleet model the average car in the fleet grew from 4.23 m in 2012 to 4.37 m in 2025. On a parallel kerb, where capacity is kerb length divided by car length plus gap, the estimates are these. θ_1 is the extra cars as a share of all stalls, θ_1^u the same per space on unmarked parallel kerb; the plan named θ_1^u its primary estimand and θ_1 its headline.

Estimate	Median	Range under the registered inputs (2.5–97.5 %)
θ_1 · all stalls, parallel bays unmarked	1.6 %	1.3 – 1.9 %
θ_1^u · per space on parallel kerb	2.7 %	2.2 – 3.2 %
stalls, on 168 424 administrative stalls	about 2 800	2 200 – 3 300
θ_1 if a quarter / half of parallel bays are painted	1.3 / 1.0 %	1.1 – 1.6 / 0.9 – 1.2 %
θ_1^u / θ_1 with a gap proportional to length	3.2 / about 1.9 %	2.6 – 3.8 % / not run separately

The fleet model fails its registered calibration check: the modelled 2025 fleet is 3.9 cm shorter than the register's, more than a quarter of the 14 cm growth that drives these estimates. A shortfall common to both years would hardly matter; one that differs between them would move the estimate, by an amount not known. Since the earlier texts had already put θ_1 near 1.6 %, its band was expected before the run, and the estimates are descriptive, not tests.

The whole range for all stalls sits in one of the bands fixed in advance. The plan's sentence for that band goes on to compare the change with the count of cars, a comparison this study could not make (see below), so only its first half is reported: on today's kerb, 2012's cars would fit between one and two and a half more cars in every hundred, under the model's inputs.

The per-parallel-space estimate, 2.2–3.2 %, straddles two bands, and for that case the plan says to give both sentences and the input that decides between them: on today's kerb, 2012's cars would fit between one and two and a half, or between two and a half and five, more cars in every hundred parallel spaces. Which of the two holds depends mainly on how long the cars made before 2010 really were and on the gap rule: with a gap proportional to length, the per-space loss sits wholly in the higher band (2.6–3.8 %). Fig. 5 shows the inputs' effect on θ_1 , not on θ_1^u .

Today's paid kerb would hold about 1.6 % more cars if they were 2012's cars.

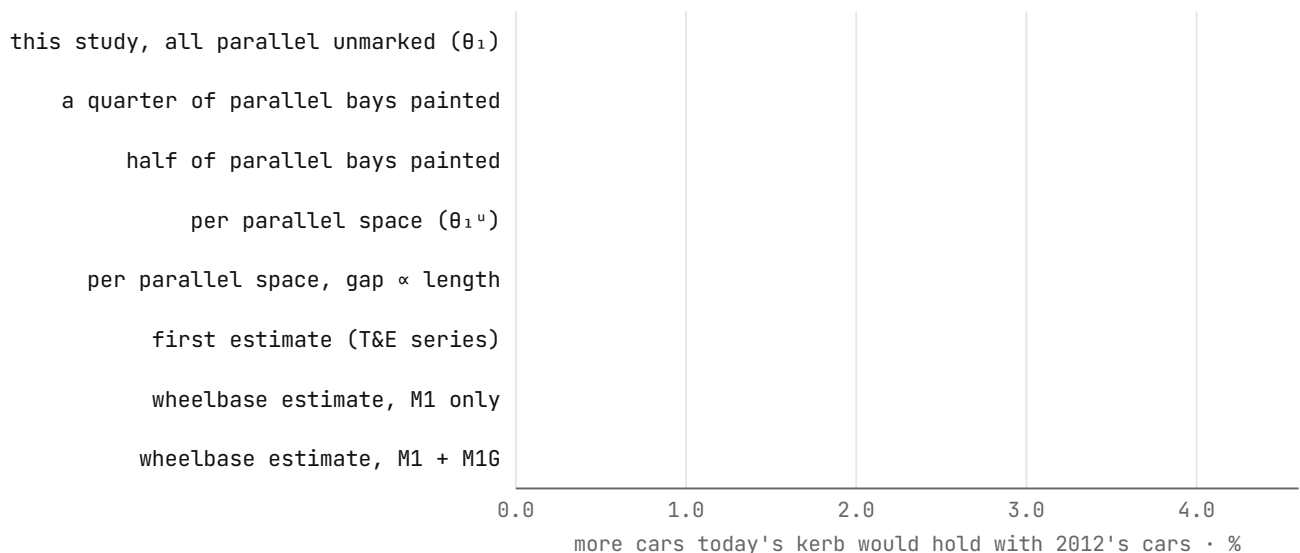


fig. 3 · How many more cars today's kerb would hold with 2012's fleet, in per cent, with the 2.5–97.5 % range under the registered input distributions: all stalls (blue) and per parallel space (black). Dashed light grey: the band edges fixed in advance. Light grey diamonds: three earlier point estimates (the first estimate, and the wheelbase estimate as first published and as corrected), for comparison only. Sources: vehicle register, export of 1 September 2026; zone layer, snapshot of 28 September 2026 (168 424 stalls).

The earlier estimates reached about the same share with a shorter fleet growth and a larger borrowed parallel share, which offset each other. Per parallel space, 2.7 % over thirteen years is below what T&E's own parameters imply (2.9–3.4 %), and far below T&E's headline pro-rated to the same span (7.4–12.1 %)¹, as the plan's sentence for that case reads; the range, 2.2–3.2 %, does overlap the first comparison. The headline itself could not be reproduced from those parameters, which give 3.4 to 4.0 % by 2040, so it is cited here as T&E's modelling and not used as a check.

Where bays are painted, a longer car does not cost a bay unless it no longer fits in one. The share of cars longer than 5.25 m, half a metre short of a painted parallel bay, rose from 0.3 % to 1.1 %. Which of Prague's bays are painted is not in the open data, so the estimate treats every parallel bay as unmarked. Painted bays could only lower it, to 1.0–1.3 % if a quarter to half are painted; under the plan's other gap rule, a gap proportional to length, which is also T&E's, it would be higher, about 1.9 %. It is therefore an upper bound only with respect to marking.

Car width and perpendicular parking

This section is descriptive and outside the registered plan. During the 2026 campaign a candidate in Praha 3 proposed switching streets from parallel to perpendicular parking¹⁸, and the two layouts ration different dimensions of the car. A parallel car uses its length plus the gap, about 5.2 m of kerb for the modelled average car of the 2025 fleet, 4.37 m (4.41 m among the register's cars in operation). A perpendicular car uses the width of its stall, 2.50 m in the Czech standard ČSN 73 6056, whose tables are quoted here from secondary sources^{19,20}. Per metre of kerb, perpendicular parking holds about twice as many cars, but its stall is 5.00 m deep against the 2.00 m width of a parallel one (both assumed from the standard's tables), so the switch turns three metres of street width, taken from the pavement or the driving lane, into stalls, before counting the wider lane a car needs to turn in. Praha 3's street widths were not measured, so how many streets have those three metres to spare is not

known. The layout also changes which kind of growth matters. On a parallel kerb a wider car is assumed to take its extra width from the lane, while in a perpendicular stall width is the binding dimension, and a new car's body, about 1.82 m wide in T&E's series for the EU, the UK and Norway¹, leaves about 68 cm between neighbours in a 2.50 m stall.

The resident permit and the city's own tariffs

A resident's permit for a first car costs 1 200 Kč a year in the main areas, 600 Kč in the outer sub-areas of Praha 5, 8, 9, 10 and 18, and 360 Kč for residents over 65. A second car costs 7 000 Kč, and since 2026 zero-emission cars pay half²¹. At 13.5 m² of net zone area per stall, the main-area price comes to 89 Kč per square metre a year (44 Kč in the outer sub-areas), whether the car is a Fabia or a Kodiaq, so the price carries no signal about the size of the car.

Because the first car pays the same whatever its length, the price per metre of kerb falls as the car gets longer. In the national fleet on the road (Prague's own fleet cannot be identified in the open data), a short car at the tenth percentile of length, 3.96 m, pays about 250 Kč per metre of kerb a year, and a long one at the ninetieth, 4.84 m, about 210. A permit priced by length, revenue-neutral (the average permit stays at 1 200 Kč), would run from about 1 140 Kč for the short car to about 1 260 for the long one, about 120 Kč a year apart (114–124 Kč across the registered range of the gap between cars), and would change the price by more than a tenth for 2 % of cars.

The level of the price is a separate quantity. A permit entitles its holder to search for a space rather than to a particular bay, and the search has its own cost²². The plan brackets the level against the city's own tariffs and never states it as one number. The first-car permit is 5 800 Kč a year below the city's price for a second car and 48 000 Kč below its fee for reserving a permanent bay, 10 Kč per m² a day²³, which pays for exclusive round-the-clock use and so marks an upper end. Both are policy prices, not market rents. The two middle points the plan named, land value at a 4 % return and garage rents, had no dated source fixed in advance and are not shown. At every price level the gap between a long and a short car is about a tenth of the price of the space itself (fig. 4).

A revenue-neutral permit priced by length would put about 120 Kč a year (114–124 Kč across the registered gap range) between a long and a short car, small next to the gap to the city's own tariffs.

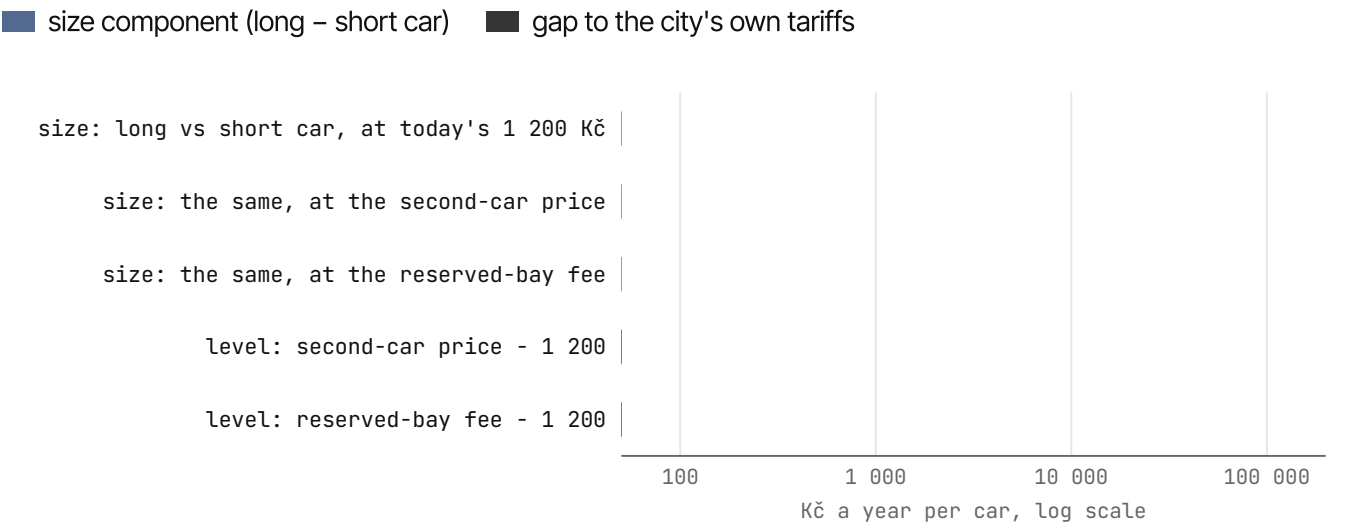


fig. 4 · Kč a year per car, log scale. Blue: the gap between a long and a short car (90th against 10th percentile of length, national fleet in operation) under a length-priced permit at three price levels. Black: the gap between the 1 200 Kč first-car permit and the city's own second-car price and reserved-bay fee. Sources: TSK price list valid 2026; Prague ordinance 16/2025; vehicle register, export of 1 September 2026.

City	Who pays	Rule
Koblenz, from March 2024 ²⁴	residents	length × width × 23.40 € a year, at least 100 €
Cologne, from March 2025 ²⁵	residents	100 € up to 4 109 mm, 110 € to 4 709 mm, 120 € above
Freiburg, 2022–2023 ²⁶	residents	240 / 360 / 480 € by length; annulled by the Federal Administrative Court in June 2023, partly because the steps between length classes were too large
Prague	residents	1 200 Kč a year for a first car in the main areas, whatever its size

Resident permits priced by car size elsewhere. Weight-based schemes (Lyon, Tübingen) and footprint schemes (Aachen, Bonn) could not be verified in primary sources and are left out, as the plan required. Prices are shown in their own currency.

The number of cars against their size

The plan's third estimate compares, in the districts whose zones barely changed since 2012 (Prague 1, 2, 3 and 7), how much more kerb the growing number of cars claims with how much more their length claims. It needs the number of valid permits at each year's end, a series that is not public. The only public figure is 215 108 valid permits on 1 May 2026, from TSK's answer 2026-05-19-058 to an information request²⁷. The series is still to be requested from TSK and the estimate was not run, so this study makes no statement about how the loss to longer cars compares with the growth in their number.

Robustness

For every input, the plan asked how far it would have to move to change the band. No single input can do so within its registered range, and the one that could, the parallel share, would have to reach 93 % or more, against the 58–60 % estimated from segment shape, an interval that reflects only the 2019 classifier's error rates, not a check on 2026 kerb.

Input (others central)	Registered range	θ_1 at the two ends
lengths of cars made before 2010	bracket of up to ± 10 cm for the oldest cars	1.30 – 1.92 %
reference pitch	4.9 – 5.5 m	1.52 – 1.71 %
mean fleet age in 2012	13.3 – 14.1 years	1.57 – 1.64 %
definition of age	± 0.5 year	1.59 – 1.63 %
shape of the survival curve (Weibull k, how sharply scrappage rises with age)	k 1.4 – 1.6	1.60 – 1.61 %
gap proportional to length, per parallel space (θ_1 , 2.7 % with a fixed gap)	registered input distributions	3.2 % (2.6 – 3.8 %)

No single input moves θ_1 out of its band; the length of cars made before 2010 matters most.

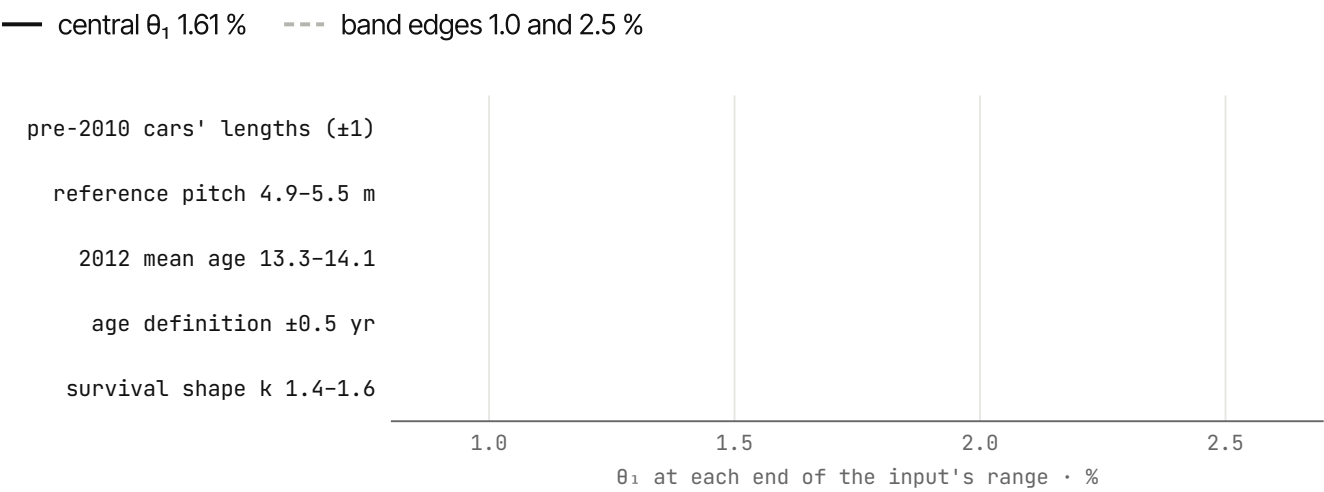


fig. 5 · θ_1 in per cent at the ends of each input's registered range, all other inputs central (blue bars), around the central 1.61 % (black). Dashed light grey: band edges at 1.0 and 2.5 %. Sources: vehicle register, export of 1 September 2026; zone layer, snapshot of 28 September 2026.

Within the registered input ranges, almost all of the spread (a first-order share of 0.92, the part of the variance one input explains alone, in a variance decomposition²⁸) comes from how long the cars made before 2010 really were. The range does not include error in the parallel share beyond the 2019 labels, the calibration gap below, or the difference between the national fleet and the cars parked in Prague. The register records them only as survivors, many of the rest were scrapped before its history begins, and the plan brackets that ignorance with a symmetric allowance of up to ± 10 cm for the oldest cars. The survival curve's shape is narrowly constrained, since, fitted to the fleet's published mean age, it reproduces the published fleet counts for 2016–2019 to within 5 % only for one narrow range of shapes (within 3 % at the shape used), and that range was fixed in the plan, before this analysis was run.

Five registered robustness checks were not run, and the ranges above do not cover those choices.

Check not run	What it varies
R4	the discrete segment model with per-segment classification
R5	the other fleet definitions, among them the Prague-registered fleet and the fleet without used imports
R6	other shapes of the age distribution
R10	the area model with width encroachment
R13	the specification curve over the gap law, the parallel share, the fleet, the segment model and the survival shape

The registered remedy for the failed calibration check, choosing the curve's shape to close the 3.9 cm gap, could not close it, because the gap is smallest at the edge of the allowed range. The shape was fixed at that edge ($k = 1.5$), and with the shape drawn across its range the estimates are the same to four decimals.

What this means

On the model, today's paid kerb would hold about 1.6 % more cars (1.3–1.9 %, range under the registered inputs) if they were 2012's cars, about 2 800 administrative stalls. The range reflects modelled inputs, mainly the lengths of cars made before 2010, not sampling error. Painted bays would lower the figure and a gap proportional to length would raise it to about 1.9 %, and the fleet model misses its 2025 calibration by 3.9 cm. New cars in Czechia grew less over 2022–2025 than T&E's series implies, though no growth at all is not ruled out. A length-priced permit would put about 120 Kč a year between a long and a short car. How the loss compares with the growth in the number of cars is not known.

What this does not show

These are counterfactuals on today's kerb, and nothing here counts the stalls Prague had in 2012. The register describes the cars registered in Czechia, while the mix parked in Prague's zones is not published, and Prague's fleet is probably younger than the national one. The lengths of the oldest cars in the 2012 fleet are assumed, with a bracket on both sides. Stalls are the city's administrative count, and a permit records an entitlement to park, not whether the car is there. The perpendicular-parking section rests on secondary sources for ČSN 73 6056. This study says nothing about how drivers would respond to a different price.

Reproduction

The analysis runs from `tools/parking6/` in the site's repository with `uv` and Python, and each script's first lines give its command; the estimates are rebuilt with `uv run --no-project --with numpy --with pandas --with pyarrow --with pyproj python tools/parking6/parking6_estimate.py`. The raw files are not in the repository; they are listed with their SHA-256 hashes in `docs/research/prague-parking-files.sha256`. The register export of 1 September 2026 (19 383 927 rows) was downloaded from the ministry's open-data site², which replaces the file as the register changes, so a later download will not match the hash. Two steps import the code of the earlier texts from a separate local repository. The order is:

1. `rsv_coverage.py`, coverage counts of the export, with no length kept;
2. `rsv_extract.py`, a compact extract of technical fields, with no VIN or owner data;
3. `parking6_design.py` all, the design-stage checks, the zone-layer snapshot and the band-width rule;
4. `eea_m1g.py` and `eea_counts_2325.py`, the EEA wheelbase series and the 2023–2025 counts;

5. `parking6_dict.py`, the model-line dictionary;
6. `parking6_estimate.py`, the tests, the wheelbase-route comparator and the estimates, written to `assets/parking6/h2.json`, `e1.json` and `e4.json`;
7. `part2_m1g.py`, the Part II correction (M1 + M1G), written to `assets/parking6/part2_correction.json`;
8. `figures.py`, the static figures.

References

1. Transport & Environment (2026). *Ever-bigger? Car size at a crossroads*, p. 26 (dimensions), pp. 29–30 (parking method). T&E report. uploads.transportenvironment.org/.../2026_06_TE_Ever_Bigger_report_final.pdf. Accessed 14 September 2026. Average dimensions of new sales, EU, UK and Norway (Inovev data), width without mirrors; 5.20 m end-to-end pitch (Berlin Traffic Information Centre, 2025); 68 % parallel share (Berlin); 8.5–14 % loss projection to 2040.

2. Shoup, D. (2005, updated 2011). *The High Cost of Free Parking*. APA Planners Press.

3. Shoup, D. (1997). The high cost of free parking. *Journal of Planning Education and Research* 17(1), 3–20.

4. Pierce, G., Shoup, D. (2013). Getting the prices right: an evaluation of pricing parking by demand in San Francisco. *Journal of the American Planning Association* 79(1), 67–81.

5. Inci, E. (2015). A review of the economics of parking. *Economics of Transportation* 4(1–2), 50–63.

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